POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

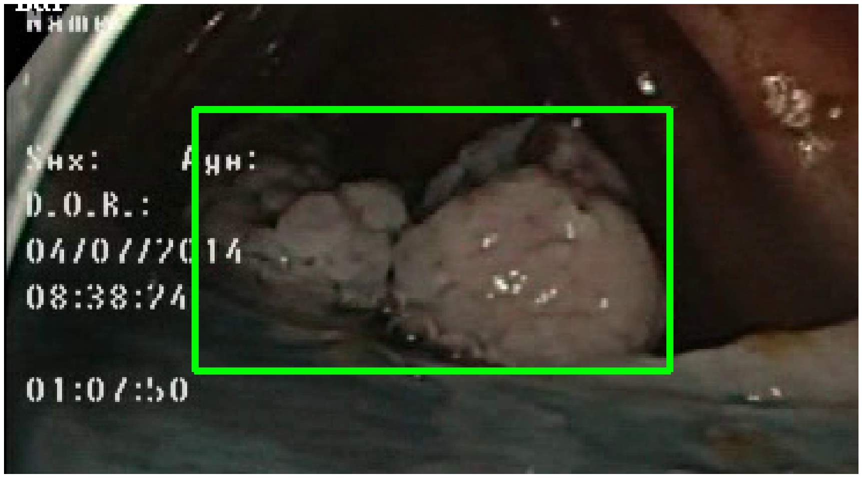
Video ID:	fdb16408-bc2c-4a65-b9d0-0ff5bfcd8b37
Total Frames:	1502
Frame Rate:	25.0 fps (assumed)
Duration:	60.1 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	273	✓
RT-DETR Detections	856	✓
MedSAM2 Detections	902	✓
Consensus (3-Model Agreement)	3	✓
Risk Classification	**Count**	**Percentage**
HIGH RISK	0	0.0%
MEDIUM RISK	1	33.3%
LOW RISK	2	66.7%
TOTAL ANALYZED	3	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — LIFTED_POLYP [LOW_RISK] (Tracked across 42 frames)

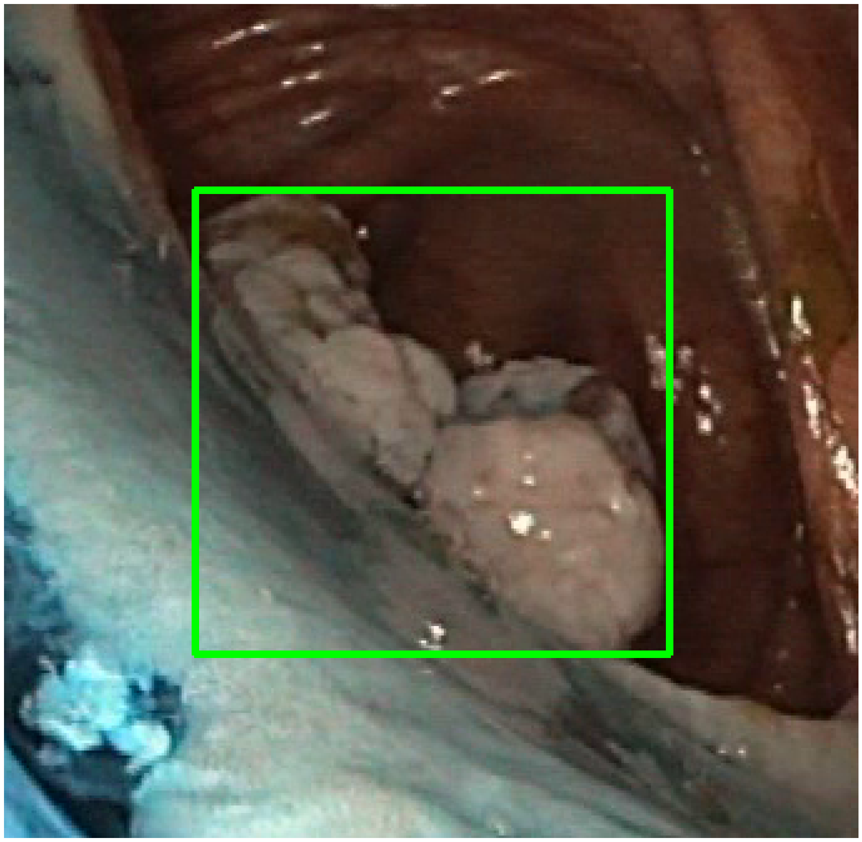


Feature	Value	Interpretation
Frame Index	910	Frame 910 of 1502
Detection Method	Detector Consensus (MedSAM2 unavailable)	YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 61.2%
Redness Score	0.046	Color-based vascularization
Relative Radius	17.5%	Polyp size as % of frame diagonal
Texture Score	0.071	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	NORMAL_MUCOSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	80.4%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #2 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 71 frames)



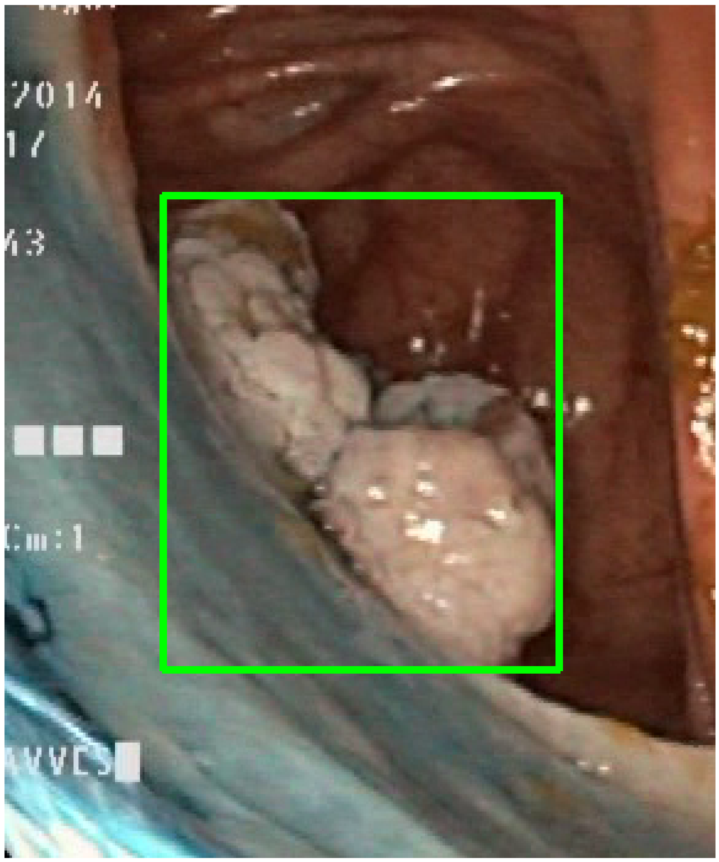
Feature	Value	Interpretation
Frame Index	789	Frame 789 of 1502
Detection Method	Detector Consensus (YOLO + RT-DETR agreement) (MedSAM2 unavailable for this video)	(MedSAM2 Unavailable)
POLYP TYPE	**FLAT_POLYP**	Type confidence: 67.8%
Redness Score	0.094	Color-based vascularization
Relative Radius	13.8%	Polyp size as % of frame diagonal
Texture Score	0.145	Surface morphology
Vessel Visibility	0.003	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output

Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	83.0%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #3 — LIFTED_POLYP [LOW_RISK] (Tracked across 91 frames)



Feature	Value	Interpretation
Frame Index	727	Frame 727 of 1502
Detection Method	Detector Consensus (YOLO + RT-DETR agreement) (MedSAM2 unavailable for this video)	(MedSAM2 Unavailable)
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 67.0%
Redness Score	0.110	Color-based vascularization
Relative Radius	31.9%	Polyp size as % of frame diagonal

Texture Score	0.083	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	NORMAL_MUCOSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	84.7%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

CLINICAL IMPRESSION

Summary:
A total of 3 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:
• HIGH RISK polyps: 0 (0.0%)
• MEDIUM RISK polyps: 1 (33.3%)
• LOW RISK polyps: 2 (66.7%)

Recommendation:
The presence of MEDIUM RISK polyps warrants clinical review and correlation with endoscopic findings.
Technical Details:
• Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
• Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
• Classification: Cluster-specific Decision Tree experts
• Confidence: Based on deep learning + medical heuristics

Disclaimer: *This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.*

AI-Based Polyp Detection System
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